US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure (TOP) 02-2-541A 4 September 2024

Safe Operation of Mobile Unmanned Ground Vehicle (UGV) Systems

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1. SCOPE.

a. This document describes the procedures for determining the mobility capabilities and characteristics of remotely and teleoperationally controlled Unmanned Ground Vehicle (UGVs). It focuses on the contribution of teleoperation to overall mobility and safe operation. This TOP does not address the operation of autonomously controlled vehicles.

b. These procedures are intended to build upon the mobility-related TOPs called out in TOP 02-2-540[1]** and to set the stage for testing autonomous mobility capabilities and characteristics. Tests of lethal and non-lethal weaponized UGVs must refer to TOP 02-2-542[2].

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement
Enclosed facility	Vehicle lab testing
Variable attenuators	± 0.5 dB
Areas with intersecting roads and narrow alleys	

2.2 Instrumentation.

Devices for Measuring	Permissible Measurement Uncertainty
Radio Frequency (RF) Resolution Bandwidth	± 15 kHz
RF Transmitted And Received Power	± .1% of value
RF Attenuation	± 1 dB
Operator Control Unit (OCU) Position	± 0.5 m
Vehicle Position and OCU	± 0.5 m
Vehicle Speed	± .4 km/hr
Vehicle Heading	± 1°
Pitch And Roll Rate	± 100°/sec
Vehicle Computer Temperature	± 1 °C
Vehicle Data Link Transceiver Temperature	± 1 °C
Vehicle Power Bus Voltage	± 1 Volt
Time	± 0.1 sec
Independent Video of Vehicle Movement	30 frames per second (fps), Infrared (IR) or color (based on test conditions).
Acceleration	± 5% of reading or 0.1 g (whichever is greater).

**Superscript numbers/letters correspond to those in Appendix B, References.

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2.2.1 Instrumentation Considerations. The selection of instrumentation and data collection equipment installed for UGV mobility testing should consider the following:

a. Equipment Mass Properties. The size and weight of the instrumentation and data recorders mounted on the test article must be small and light in comparison to the test article. The use of disproportionately large or heavy equipment may have an unacceptable effect on test results. Where feasible, weight should be removed to keep mass properties (center of gravity, weight, etc.) constant when an instrumentation package is added.

b. Real-Time Data Downloading. Wireless transmission of test data may interfere with normal system operation due to electromagnetic interference. Prior to test execution, compatibility between real-time data downlinks and system operation must be verified.

c. Spectrum Analyzers. This equipment is needed to identify the presence of other frequencies in use at any given time in the test and should be operated in the vicinity of the UGV and the OCU. Although systems are required to have a frequency authorization prior to the start of testing, establishing cause and effect during UGV mobility testing requires a means of ensuring that frequency interference is not a factor in test results.

d. UGV System Data Bus. A significant source of data in a test effort relative to UGV mobility will come from the UGV system data bus, especially for advanced autonomous capabilities. This information is useful for establishing cause and effect when examining test results, as well as for monitoring the test article to ensure safety on the test range. System data taps and supporting data harvesting and analysis tools must be compliant with the data bus standards of the test article. Many emerging UGV systems are compliant with the Joint Architecture for Unmanned Systems (JAUS) standard. The Joint Ground Robotics Enterprise (JGRE), formerly known as the Joint Robotics Program, established JAUS as a standard architecture for UGVs developed within the JGRE.

3. REQUIRED TEST CONDITIONS.

a. UGV operators must meet training requirements for system operation. Operators, independent emergency shutdown system operators, bystanders, and support personnel must be familiar with the test area, potential hazards, and relevant Standing Operating Procedures (SOPs).

b. Software testing must be sufficiently completed as defined in TOP 02-2-540 and must be completed in accordance with MIL-STD-882E[3] and the Joint Services – Software Safety Authorities (JS-SSA) Software System Safety Implementation Guide[4].

c. The test article must comply with the safety and instrumentation requirements defined in TOP 02-2-540, including the following:

(1) For any UGV capable of injuring or killing a person, an independent emergency shutdown system must be properly installed, and its functionality must be verified prior to any other testing or training activities. This can be comprised of a manual push button emergency

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shutdown system located at in easily accessible location, or a remote activated emergency shutdown system that maintains positive control. If it is a remote emergency shutdown system, it should be tested first while the UGV under stationary non-active state, then tested while the UGV is active under a teleoperated state.

(2) UGVs must be instrumented to support the critical information analysis and displays as needed by the test officer to support real-time test control and monitoring.

(3) Unique warning lights that warn nearby personnel of vehicle movement must be installed. It must be verified that operation of these lights does not interfere in any way (physically or electronically) with the system under test (SUT).

(4) Lift and tie-down procedures, as well as procedures for loading the SUT onto a transport vehicle must be finalized prior to the start of testing.

d. Standoff distances from the UGV must be established for personnel and equipment in forward, lateral, and rearward directions. Standoffs must consider where the vehicle could go during uncommanded movement and full throttle acceleration. Blind spots should also be considered in the decision of where to place personnel and equipment.

4. TEST PROCEDURES.

4.1 Vehicle Subsystem Tests.

These tests identify key operating characteristics associated with UGV subsystems. Completing these tests first in a UGV test program are part of an overall strategy to slowly build confidence in UGV system robustness and to prepare the tester for additional phases of testing.

4.1.1 Tethered Controller Test.

This test identifies the operating characteristics of UGVs while under control of a hand-held controller that plugs directly into the vehicle. Such a device is typically used for vehicle movement within maintenance facilities, for loading or unloading the vehicle from a trailer, subterranean operations, or anytime vehicle movement on a test range cannot be performed via RF data link. It also tests for lockouts that prevent other controllers from taking control of the vehicle. For this phase of testing, the system should be exercised with the vehicle tracks/wheels suspended above ground (e.g., on jack stands) before performing full-function checks with the unrestrained vehicle.

a. Ensure the OCU for RF control of the UGV is powered off.

b. Perform the Vehicle Fine Control subtest (paragraph 4.4.3) using the tethered controller.

c. Power up the OCU and look for indications that the tethered controller has control of the vehicle.

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d. Use the OCU to attempt to take control of the vehicle. Record indications of controlling the unit before and after the OCU control attempt.

e. Disconnect the tethered controller and allow OCU to take control of the vehicle.

f. Plug tethered controller back into the vehicle and attempt to control vehicle. Record observations that indicate the tethered controller now controls the vehicle.

4.1.2 System Field-of-View (FOV) Test.

This test identifies regions around the vehicle that cannot be seen by the remote operator via teleoperation, and identifies the effort required to detect people or impairments to mobility near the vehicle.

a. Use the procedures outlined in TOP 03-2-812[5] to obtain a field of vision of sensors as presented through the OCU. Be sure to address each extreme in sensor pan, tilt, and FOV setting. Measure the FOV of each sensor at each range setting.

b. Record the list of operator actions and timelines required to use one sensor to observe a blind spot of another sensor.

c. Using collected data, identify residual blind spots. Residual blind spots are areas that cannot be observed by switching between views or by combining views.

d. Calculate the size and location of residual blind spots. Determine if people or objects large enough to pose safety risks or impair mobility could fit within residual blind spots.

4.1.3 Vehicle State Reporting Test.

This test is used to identify variations between the reported and actual vehicle states. Accurate state reporting is essential for ensuring that the vehicle is safe enough to approach and handle, and for performing certain tasks according to the intended mission.

a. Identify key vehicle data reported to the OCU. States needed for safe operation during mission execution include position, speed, heading, steering, pitch and roll attitude, and reporting sensor viewing angles. States needed to ensure that the vehicle is safe to approach include gear and brake settings, engine power level, vehicle electrical power state, and the states of any articulating parts (e.g.: turrets, arms, etc.).

b. Record both the OCU reported vehicle data and the vehicle truth data. Obtain truth data from test instrumentation independent of the system and from visual observation of the system.

c. Identify variations between reported and actual states that may affect safe vehicle operation.

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4.2 Emergency Shutdown Test.

This test provides confidence to the testers that the emergency shutdown system can be relied upon to shut the UGV down in a manner that enables it to be safely approached. This subtest is intended for both RF and vehicle-mounted shutdown systems. Of particular importance is the testing of independent emergency shutdown systems that may be applied to some systems. Conduct in conjunction with the Data Link Degradation Effects subtest (paragraph 4.3) with the following additions:

a. Identify each available means of shutting down a vehicle in an emergency on a test range should the vehicle become uncontrollable. These may include test center applied independent stop systems, prime item developer independent stop systems, or any other system that is not intended to be a part of the UGV system and that does not use the system's data link. Systems must maintain positive control and communication to the platform to be utilized.

b. Restrain or confine the vehicle, then power it up and establish a slow wide turn. Wheels or tracks should be turning as the vehicle is steered to negotiate a large radius turn.

c. Use an emergency shutdown system to shut down the vehicle. Record the results of each shutdown.

d. Repeat for each means of shutting down the system in an emergency.

4.3 Data Link Degradation Effects.

This test identifies the effects of data link degradation on overall system performance and operating characteristics.

a. Position the UGV on jackstands or similar restraint to prohibit dangerous movement. The operator and the OCU should remain nearby the UGV initially to provide a reliable RF connection. Alternatively, in a laboratory setting, hardwire the OCU and the vehicle together with variable attenuators between them, allowing for simulation of communication distance.

b. If available through system software or documentation, note the time period the UGV is to wait before executing lost link procedures. If available, select a lost link behavior with discernable changes in vehicle function (e.g.: hard turn, back up, etc.).

c. Operate the vehicle while it is restrained or elevated off of the ground according to a defined set of planned functions.

d. Incrementally increase the RF attenuation by increasing distance between the OCU and UGV and repeat vehicle operation. A spotter must remain with the restrained UGV to observe and record system behavior.

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e. As affects become apparent record the distance and observations on the effects of data link degradation on both vehicle operation and OCU imagery. Record evidence of degradation effects

f. Continue until data link is lost.

g. Record observations on UGV actions and OCU indications when data link is lost.

h. Allow vehicle to remain in lost link state for 3 minutes or 20% longer than what was programmed for lost link procedures, whichever is longer. This allows time for any undocumented vehicle behaviors to become apparent.

i. Decrease the distance by half the amount by which it was being increased and wait 30 seconds for data link reacquisition to occur. Repeat this step until link is reestablished.

j. Record observations on UGV actions and OCU indications when data link is restored.

4.4 Automotive Performance.

These tests identify overall system performance under each mode of control.

4.4.1 Tailored Automotive Tests.

Conduct automotive tests as defined in TOP 02-2-540. For each of the following subtests include the following information as appropriate to intended mission:

a. Grades and Slopes. Record the information made available to the remote operator regarding slope angle and impending vehicle roll over. Record remote operator indications of nose slide or tail slide. Record the grade holding ability while halted, while the emergency stop is engaged, and while powered off. Conduct for each mode of control.

b. Standard Obstacles. For each mode of control, maneuver the vehicle into position and conduct one trial of negotiating the standard obstacles (stairs, curbs, vertical wall, V-ditch, gap crossing, etc.). Record observations on OCU control inputs required for steering effort. Record issues associated with maneuvering the vehicle up to or over the obstacles.

c. Fording. Identify effects of water near or on driving sensors to include degradation of imagery at the OCU. Record indications of loss of controllability or traction available to the remote operator. Record difficulties with water entry or exit associated with each mode of control. Record any water penetration into the vehicle and components affected.

d. Soft Soil Mobility. Record indications of loss of controllability or traction available to the remote operator for each terrain tested. Record the relevant characteristics of the terrain, such as soil moisture content, grass height, snow depth, etc.

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e. Steering and Handling. For each mode of control, record steering and heading indications available to the remote operator. Record observations on suitability of the information available to the remote operator (steering and handling indications and the OCU FOVs) to support steering-and-handling-related functions. Record observations on OCU control inputs required for steering effort.

f. Acceleration/Maximum Speed. Conduct for each mode of control and terrain type. Record observations of OCU indications of vehicle speed and acceleration.

g. Braking. Conduct for each mode of control. Record observations of OCU indications of vehicle speed and braking effort.

4.4.2 Test for Driver Assist Functions.

Driver assist functions relieve the operator of certain burdens in operating the vehicle remotely or offer a degree of vehicle protection against operator actions that could inadvertently impair vehicle motion. They DO NOT relieve the remote operator of the fundamental task of driving the vehicle. Examples include speed control, heading control, and autonomous rollover avoidance to correct operator inputs that would otherwise cause a rollover.

a. During automotive tests described in para 4.1.1, verify that known driver assist functions can be activated and deactivated as needed and that they function as intended.

b. Record OCU indications of the state of driver assist functions.

c. Record observations on system operation with driver assist function on and off.

d. Over the course of the overall test program, record additional observations that may imply the presence of undocumented driver assist capabilities.

4.4.3 Vehicle Fine Control.

Although much of the conventional automotive testing examines the limits of automotive performance, this test is intended to examine the fine control inputs and responses needed for some situations. The ability to tap and record operator inputs (joystick angle, pedal angle, speed control, etc.) and vehicle speed and acceleration is required to accurately characterize these behaviors. Testing should be performed on a flat, paved surface. This test may be combined with the Tailored Automotive Tests (paragraph 4.4.1).

a. Speed and Acceleration.

(1) For each mode of control, instruct the remote operator to drive at set speeds throughout the available speed range. Record instrumented speed input and vehicle speed during each trial to determine the ability of an operator to finely control speed.

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(2) For each mode of control, instruct the remote operator to accelerate from minimum to maximum vehicle speed using acceleration inputs available throughout the available range. Record instrumented speed input and vehicle acceleration during each trial to determine the ability of an operator to finely control acceleration.

(3) For UGVs with adjustable speed settings, record the minimum step of speed adjustment that can be set by the operator.

b. Braking. For each mode of control, instruct the remote operator to decelerate from maximum to minimum vehicle speed using braking inputs available throughout the available range. Record instrumented brake input and instrumented vehicle deceleration during each trial to determine the ability of an operator to finely control deceleration.

c. Steering.

(1) For each mode of control, instruct the remote operator to steer at set angles throughout the available steering and vehicle speed ranges. Record instrumented steer input and resulting vehicle position during each trial to determine the ability of an operator to finely control steering.

(2) For UGVs with adjustable steering speed settings, record the minimum step of steering speed adjustment that can be set by the operator.

4.5 Teleoperation Effectiveness.

This test examines how well the overall operating characteristics of the UGV work together to support the soldier in employing a teleoperated UGV.

4.5.1 OCU Usefulness.

While many teleoperated systems have similar components and displays, significant variations may exist in the ease with which the UGV can be operated. Human Factors Engineering testing should be performed according to TOP 01-2-610[6] to determine if screen size, brightness, control spacing, and control size are appropriate

4.5.2 Corner Negotiation.

This test is intended to identify UGV safe operating limitations due to difficulties the remote operator may have in seeing the area into which the vehicle is about to turn or in maneuvering the vehicle into a narrow area.

a. Beginning at off-road speed, slow the vehicle as needed to make turns onto intersecting roads. Two of the intersecting roads must be narrow: one road must be approximately 75% wider than the UGV, and one must be approximately 50% wider than the UGV. Turns must be made onto the narrow intersecting roads from both the left and the right.

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For UGVs designed for low-light operations, turns must be conducted in both daytime and in nighttime conditions.

b. Conduct at least 3 trials of each condition.

c. Observe the remote operator's actions needed to clear the area before making the turn. Note the timelines associated with switching OCU displays between looking ahead of the vehicle and looking into the turn.

d. Collect comments from remote operators and independent observers on the suitability of teleoperation capabilities to safely turn the UGV.

4.5.3 Latency.

TOP 02-2-546[7] details procedures for measuring latency of teleoperated UGVs.

a. The procedures for basic video latency measures the lag, or time between a real-time event and the corresponding video display of that event on the OCU screen.

b. Command-to-action latency measures the time between an operator input and a corresponding system output. This could include motion commands, gripper movements, or subsystems like lights.

c. End-to-end system latency combines the two others, measuring the time between an operator input, the corresponding system output, and the display of that output on an OCU screen.

4.6 Environmental Operating Condition Tests.

These tests are intended to identify operating characteristics of teleoperated UGVs that result from exposure to extremes in environmental conditions.

a. Select environmental tests as defined in TOP 02-2-540 based on system requirements documents, intended missions, and mission locations.

b. Conduct Vehicle Fine Control (paragraph 4.4.3) and Teleoperation Effectiveness (paragraph 4.5) subtests, to the maximum extent possible, while exposed to required environmental conditions. If operators or other support personnel are exposed to extreme environmental conditions, consider their safety and effectiveness in their roles for each scenario.

c. Record the environmental effects on overall system performance and safety. For key changes in system performance, specifically address the environmental effects that caused the performance reduction.

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4.7 System Anomalies.

This section outlines additional information to be collected during the test program.

a. Early in the test effort, identify OCU indications of abnormal vehicle operation, and assess the safety status of the vehicle during each problem.

b. During each anomaly encountered in the test effort, make note of the remote operator's ability to determine that a problem exists and to understand the nature of the problem.

c. Document all problems that involve the vehicle being in a state that is unsafe to approach.

d. Identify general recommended improvements to system documentation based on observed system capabilities and characteristics.

5. DATA REQUIRED.

5.1 Vehicle Subsystem Tests.

5.1.1 Tethered Controller Test Data.

a. Data from Vehicle Fine Control test.

b. List of indications at OCU that tethered controller has taken control of vehicle.

c. List of indications at OCU that OCU has taken control from tethered controller.

d. OCU can take and retain control of vehicle during tethered control (Y/N).

e. Tethered controller can take and retain control while under OCU teleoperation control (Y/N).

5.1.2 Vehicle Blind Spot Test Data.

a. Field of view (degrees horizontal, degrees vertical) for each setting of each sensor at each extreme in pan and tilt.

b. Azimuth and elevation boundaries of residual blind spots for areas that cannot be observed with any combination of sensors selectable at the OCU.

c. Size and locations of blind spots.

d. List of operator actions required to observe single sensor blind spots.

e. Time required to use one sensor to observe the blind spot of another sensor (sec).

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(1) Sensor identification, pan and tilt setting, and FOV setting that has a blind spot.

(2) Sensor identification, pan and tilt setting, and FOV setting that is used to observe the blind spot.

f. Residual blind spot assessment:

(1) A person can fit in the residual blind spots while crouched (Y/N).

(2) The residual blind spots can contain an object large enough to prevent vehicle motion (Y/N).

(3) The remote operator can see enough via the OCU to maneuver the vehicle in confined areas (Y/N).

5.1.3 Vehicle State Reporting Test Data (Reported and Truth).

a. Orientation and motion states:

(1) Heading (deg).

(2) Pitch and roll angles (deg).

(3) Speed (km/hr).

(4) Direction of travel (forward, reverse).

(5) Position.

b. Actuator commanded and reported states:

(1) Throttle (%).

(2) Braking (%).

(3) Steering (deg from centerline, direction of deflection).

(4) Gear (gear id).

(5) Vehicle electrical power.

(6) States of articulating parts (locked, unlocked, etc.).

5.2 Emergency Shutdown Test Data.

List of indicators of Emergency Shutdown System degradation.

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5.3 Data Link Degradation Effects.

a. Lost link mission plan settings.

b. Response and controllability data.

(1) Time command was issued by operator (sec).

(2) Time vehicle began to respond (sec).

c. Observed Effects.

(1) Screen captures or video captures of OCU imagery showing each stage of imagery degradation.

(2) List of attenuation settings and noted observations on OCU display quality and vehicle controllability.

(3) Observations on changes in system performance.

5.4 Automotive Performance.

5.4.1 Tailored Automotive Tests.

Supplemental data requirements for automotive performance TOPs include the following:

a. Grades and Slopes.

(1) Mode of control.

(2) OCU indications of slope angle, pitch and roll, and impending roll over.

(3) OCU indications of nose slide or tail slide.

(4) Observations on suitability of OCU displays for operation on slopes and grades.

(5) Grade holding ability while halted, emergency stopped, and while powered off.

b. Standard Obstacles.

(1) Mode of control.

(2) Dimensions and characterization of the obstacle.

(3) Time elapsed from first contact of vehicle with an obstacle to when the obstacle was completely passed (sec).

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(4) List of extraordinary remote operator actions (actions other than those needed for normal vehicle operation) required to negotiate the obstacles.

(5) Observations of issues associated with maneuvering the vehicle up to or over obstacles.

c. Fording.

(1) Mode of control.

(2) List of effects of water near or on driving sensors to include degradation of imagery at the OCU or error states.

(3) List of OCU indications of loss of traction.

(4) Observations on difficulties with water entry or exit.

(5) Description of any water intrusion into the vehicle.

d. Soft Soil Mobility.

(1) Mode of control.

(2) List of OCU indications of loss of traction.

(3) Characterization of the terrain.

e. Steering and Handling.

(1) Mode of control.

(2) List of OCU steering and heading indications.

(3) Observations on information available to the remote operator (steering and heading indication and OCU FOVs) to maneuver the vehicle.

f. Acceleration/Maximum Speed:

(1) Mode of control.

(2) Terrain type.

(3) Commanded speed.

(4) Achieved steady-state speed.

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(5) Observations on the system's ability to finely adjust acceleration and speed.

g. Braking.

(1) Mode of control.

(2) Commanded speed.

(3) Achieved steady-state speed.

(4) Observations on the system's ability to finely adjust deceleration and speed.

(5) Description of marking used to identify the intended stopping location.

(6) Observations on the system's ability to brake to a designated spot.

5.4.2 Test for Driver Assist Capabilities.

a. Descriptions of observed driver assist capabilities.

b. Test conditions under which driver assist functions were observed.

c. OCU indications of driver assist status.

d. Remote operator can engage and disengage driver assist functions as needed (Y/N).

e. Driver assist is only "on" with remote operator's knowledge (Y/N).

f. Observations of driver assist performance.

5.4.3 Vehicle Fine Control.

a. Maximum speed: meters per second.

b. Speed Resolution: Number of speed variations available, or "Continuously Variable" if a very high number.

c. Acceleration response: time to accelerate from zero to top speed (sec).

d. Decelerate response: time to slow to zero using only throttle control (sec).

e. Brake resolution: able to incrementally adjust braking effort (Y/N).

f. Brake response: distance covered when stopping from full speed to zero (m).

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g. Steering resolution: minimum steering input step size, or “Continuously Variable” if a very high number.

h. Steering response time: time required to go from straight ahead to full deflection in one direction (sec).

i. Observations on ability of the remote operator to maintain specified courses, slow speeds, and to brake and accelerate gently.

j. The minimum UGV acceleration, steering, and braking response that can be commanded by an OCU operator.5.5 Teleoperation Effectiveness.

5.5.1 OCU Usefulness.

a. Assessment of overall image quality, controls, and vehicle states to help the remote operator maneuver the vehicle around obstacles and through tight areas.

b. Description of the situations in which remote operator did not understand what was happening at the vehicle.

c. Screen captures or photos highlighting information supplied to remote operator while maneuvering the vehicle.

5.5.2 Corner Negotiation.

a. List of actions required for the remote operator to observe the area into which the vehicle is about to turn.

b. Observations on the success rate of completing turns into narrow paths.

c. Observations on the ability of the remote operator to look in one direction using vehicle sensors while the vehicle is moving in another.

d. Observations on the time and the care required by the remote operator to make turns.

5.5.3 Latency.

a. Radio configuration and frequency settings.

b. List of equipment used to capture latency measurements, and its applicable configuration settings.

c. Results of each latency trial.

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5.6 Environmental Operating Condition Test Data Requirements.

a. Data as defined in environmental TOPs and standards.

b. Data as defined in Vehicle Fine Control and Corner Negotiation subtests.

c. List of effects on OCU: menu responsiveness, control freedom of movement, screen resolution, brightness, contrast, and overall image quality for each sensor FOV and setting.

d. List of effects on the vehicle: vehicle responsiveness, vehicle fine control, and controllability.

e. Observations on ability of the OCU to be used to maneuver the vehicle safely and effectively.

f. Assessment of video display to support full vehicle mobility at required speeds during bright sunlight and in total darkness.

5.7 System Anomalies.

a. List of uncommanded or unresponsive situations.

(1) Occurrences of incorrect status reporting.

(2) Occurrences of uncommanded movement or state changes.

(3) Occurrences of violating safe boundaries in time or in space.

(4) Occurrences of abrupt system behaviors.

b. Observations regarding recurring anomalies (actions leading up to anomaly, environments, etc.).

c. Anomaly Resolution.

(1) Observations on sufficiency of information presented to the remote operator to identify anomalies.

(2) Observations on ability of the remote operator to remotely correct and resume mission.

(3) Data from vehicle data tap.

(4) Data from OCU data tap.

d. List of anomalies causing unsafe conditions.

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e. Training and Documentation Improvements.

(1) Remote operator comments on system operation and mobility.

(2) List of discrepancies between actual vehicle operation and available documentation.

(3) List of inadvertent operator errors while maneuvering and controlling the vehicle.

6. PRESENTATION OF DATA.

6.1 Vehicle Subsystem Test Results.

6.1.1 Tethered Controller Test Results.

a. Present a table showing maximum automotive performance demonstrated during tethered controller operation for speed, steering, and braking.

b. Describe the automotive characteristics of the UGV under tethered control in context of vehicle operation in facilities. Describe control responsiveness and sensitivity to throttle, steering, and braking commands to enable both smooth changes for normal operation as well as abrupt changes needed to suddenly avoid a collision. Provide a recommendation for the number of people needed to act as safety observers while under tethered control to protect nearby personnel, facility equipment, and the test article.

c. Describe lockouts that prevent OCU control during tethered control as well as order of precedence available when a tethered controller is activated during OCU control.

d. Describe the suitability of the tethered controller for test support activities such as maneuvering the vehicle into and out of facilities, on and off ramps for transporting the vehicle, and getting the vehicle out of impaired mobility situations.

6.1.2 Vehicle Blind Spot Test Results.

a. Present side and top-down views of the UGV showing coverage of each sensor aboard the vehicle.

b. For systems that enable combined views or that enable the remote operator to shift between views, present one additional set of views (side and top-down) showing sensor coverage areas superimposed.

c. Describe the size of the residual blind spots that cannot be observed by switching or combining views that could contain a person (standing or crouched).

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d. Describe the operator actions needed to detect people and impairments to mobility (e.g.: posts, trees) near the vehicle, such as changing sensor selections or settings during system operation. Describe the consequences of those required actions on safe operation of the system around people, equipment, and facilities.

6.1.3 Vehicle State Reporting Test Results.

a. Describe the ability of the vehicle to promptly and accurately report its state to the OCU in the areas of: vehicle orientation and motion information such as heading, pitch and roll attitude, speed, and direction of travel; commanded and reported automotive control settings for throttle, brake, steering, and gearing; and driving sensor state information such as sensor selection, sensor FOV, sensor orientation settings such as pan and tilt, and sensor discrimination settings such as white-hot/black-hot.

b. Describe the suitability of reported information to safely operate the vehicle.

c. Describe the suitability of reported information to know when the vehicle is safe to approach.

6.2 Emergency Shutdown Test Results.

a. Describe indicators of data link degradation on each available emergency shutdown system.

b. Describe the overall effectiveness for each available emergency shutdown system. Describe the effects of emergency shutdown system activation on the vehicle along with the timeliness of each of those effects. Also, describe sufficiency of available indicators to the remote operator and to the independent shutdown system operator that the system is safe to approach and handle.

6.3 Data Link Degradation Effects.

a. Present a table or graph showing data link attenuation at each major change in system responsiveness or change in video quality sufficient to affect teleoperation of vehicle movement.

b. Describe the observed effects of data link degradation on system responsiveness and OCU imagery. Address changes in system response timeliness and the rate at which this degradation occurs (rapid, very gradually, etc.). Describe how changes in OCU imagery affect the ability of a remote operator to maneuver the vehicle safely. Include screen captures and associated conditions as needed to help describe the relationship between data link degradation and overall controllability and mobility of the system.

6.4 Automotive Performance.

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6.4.1 Tailored Automotive Tests.

Present the results of each automotive test as required by each automotive TOP under each mode of control along with supplemental data defined in paragraph 5.4.1.

6.4.2 Test for Driver Assist Functions.

Describe any observed driver assist capabilities such as autonomous rollover avoidance actions, heading control, or speed control. Describe ability of remote operator to engage, disengage, and observe current state of any driver assist functions. Describe the observed functionality of each driver assist function.

6.4.3 Vehicle Fine Control.

Describe overall ability of the system to apply automotive capabilities in each mode of control. Include a narrative on the ability of the vehicle to be lined up properly to traverse obstacles in a timely manner, to be stopped at intended points, and to be turned at intended locations.

6.5 Teleoperation Effectiveness.

6.5.1 OCU Usefulness.

Describe the usefulness of vehicle controls, states, and imagery to support teleoperation.

6.5.2 Corner Negotiation.

Describe the remote operator's actions needed to clear the area into which the vehicle will make sharp turns. Describe the extent to which the operator can safely clear the area into which the vehicle will turn while ensuring the area is clear ahead of the vehicle. Describe safety implications of demonstrated corner negotiation capabilities and limitations.

6.6 Environmental Operating Condition Testing.

a. Describe the environmental effects on Vehicle Fine Control and Teleoperation Effectiveness.

b. Describe the environmental effects on OCU controls. Describe the effects on display responsiveness, brightness, contrast, and resolution.

c. Describe the environmental effects on vehicle responsiveness and controllability.

d. Describe the environmental effects on data link performance.

e. Summarize the overall environmental effects on system performance and safety.

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6.7 System Anomalies.

a. Present a table summarizing uncommanded activities and unresponsive situations with associated conditions.

b. Clarify or highlight specific abnormalities as needed and cite Test Incident Reports (TIRs), video clips, or other sources if useful.

c. Describe the sufficiency of information available to the operator to remotely identify and correct anomalies and continue with a task. Identify anomalies that the system can correct on its own (i.e.: do not involve remote operator actions) and enable task continuation without reprogramming or re-initialization.

d. List the anomalies that leave the vehicle in an unsafe state to approach and require subsequent activation of the emergency shutdown system.

e. Describe the areas where training or documentation needs emphasis to ensure safe operation of the system.

f. Provide simple recommendations for system improvement.

6.8. Test Data Examples.

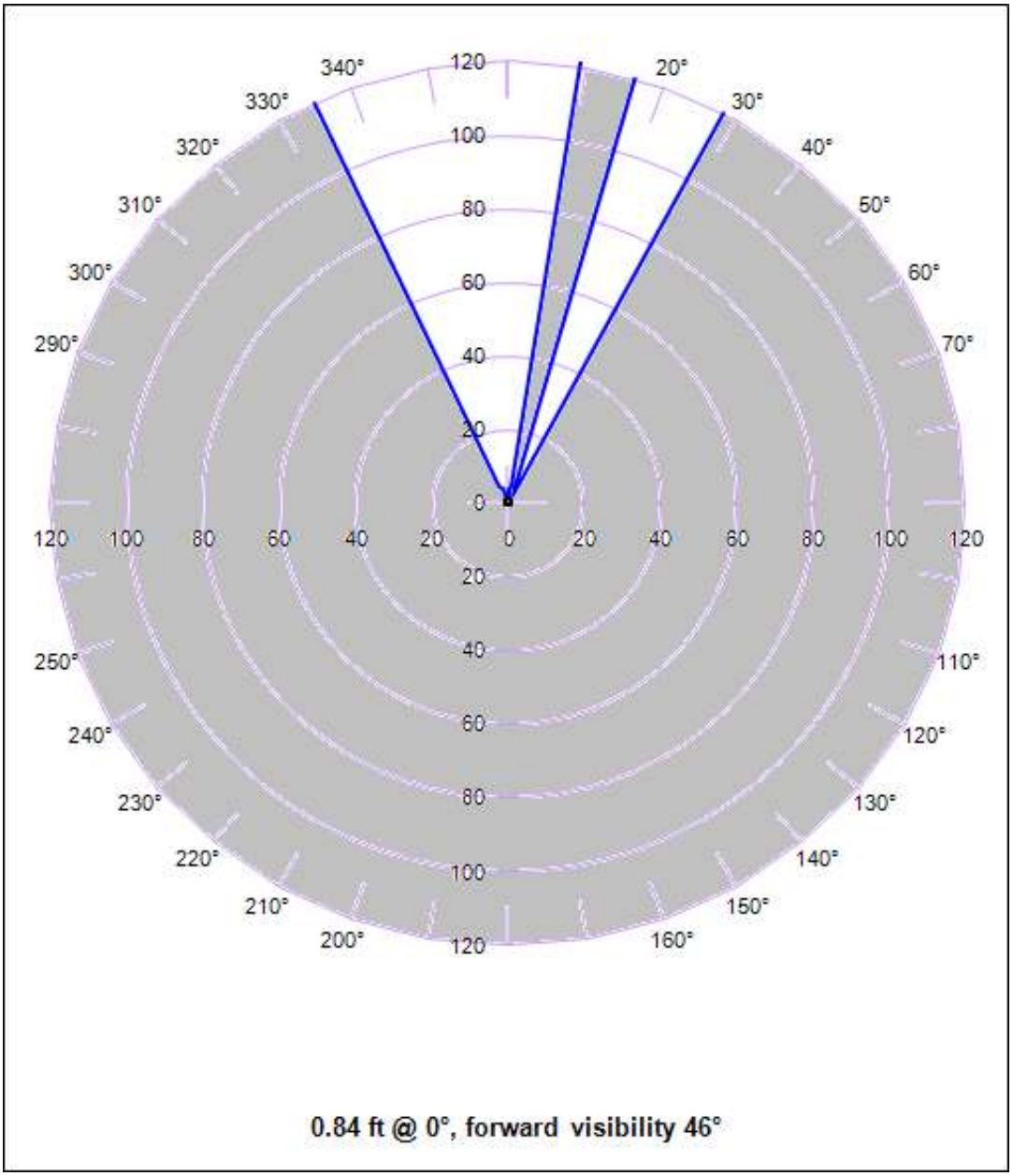
6.8.1 Vehicle Blind Spot Test Data Example.

a. Camera zoom and Field of View (FOV). FOV testing was conducted in accordance with MIL-STD-1472H, paragraph 5.12.5.1. FOV measurements of the Xxxx were taken with the Xxxx in the xxxx pose and with the xxxx in the up position. FOV measurements were taken through the xxx, xxx, xxx, and xxx cameras. Measurements were taken to determine the overall FOV for each camera. Data were collected on a flat level surface with the cameras positioned over the center of the turning circle. Data points were recorded in 10° increments to define FOV.

(1) Xxxx camera, xxx pose. The FOV for the xxxx camera in xxxx pose is illustrated in Figure 1. The forward visual field of view for the xxxx camera in xxxx pose was 83°. The minimum ground view at all distances beyond 3 m (10 ft) at 0° was 0.33 m (1.09 ft), which met the requirement according to MIL-STD-1472F, paragraph 5.12.5.2. The obstructions between 329° and 340° and 14° and 24° were because the xxxx were in the up position.

(2) Xxxx Camera, xxxx pose. The FOV for the Xxxx Camera in xxxx pose is illustrated in Figure 2. The forward visual field of view for the Xxxx Camera in xxxx pose was 46°. The minimum ground view at all distances beyond 3 m (10 ft) at 0° was 0.26 m (0.84 ft), which met the requirements according to MIL-STD-1472F, paragraph 5.12.5.2. The obstruction between 9° and 16° was because the flipper on the right side was in the up position.

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Note: Ground distance measured in meters; the shaded area indicates obstructed or non-visible view.

Figure 2. FOV from xxxx camera, xxxx pose, Xxxx.

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6.8.2 Standard Obstacles Data Example.

a. Trench Crossing.

(1) The Xxxx was operated across a “V’ profile to determine if any points of contact other than the track make contact with the ground surface. Figure 5 displays the V-ditch obstacle used for this test. Figure 6 shows the dimensions of the obstacle.



Figure 3. V-Ditch.

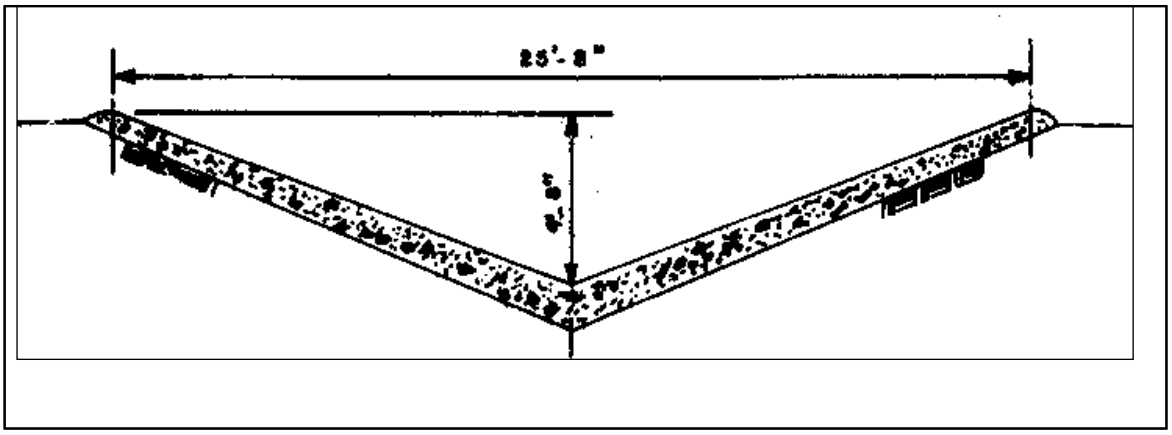


Figure 4. Cross-Section of V-Ditch.

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(2) The Xxxx was operated to the edge of the V-ditch, and was slowly operated across the obstacle. Upon descending the V-ditch, it was recorded that the nose of the Xxxx made contact with the opposite side of the V-ditch causing minor damage. After initial contact, the Xxxx was successfully able to traverse the V-ditch and ascend to the other side.

6.8.3 Acceleration/Maximum Speed Data Example.

a. Forward Direction. The vehicle was driven in a straight line from a standstill position up to maximum forward velocity with the OCU set to the X throttle setting. A minimum of X acceleration runs was conducted with X runs performed in each test course direction. The test results obtained from the X runs were then averaged to determine the maximum speed. Maximum vehicle speed was defined as the speed limited either by governor actuation, available vehicle power, or, by software limitation.

TABLE 1. FORWARD DIRECTION, GRAVEL

Selected Speed	Test Run Speed (mph)					Average Speed (mph)
	1	2	3	4	5	
XXXX	0.60	0.61	0.60	0.60	0.60	0.60
XXXX	3.05	2.98	3.03	3.05	2.99	3.02
XXXX	5.90	5.98	5.90	6.14	6.06	6.00

TABLE 2. FORWARD DIRECTION, GRASS

Selected Speed	Test Run Speed (mph)					Average Speed (mph)
	1	2	3	4	5	
XXXX	0.59	0.59	0.59	0.59	0.59	0.59
XXXX	2.97	2.99	2.91	2.89	2.91	2.93
XXXX	5.98	5.98	5.98	5.90	5.68	5.90

TABLE 3. FORWARD DIRECTION, PAVED

Selected Speed	Test Run Speed (mph)					Average Speed (mph)
	1	2	3	4	5	
XXXX	0.60	0.59	0.59	0.58	0.59	0.59
XXXX	2.98	2.96	2.97	2.95	2.98	2.97
XXXX	5.88	5.98	6.06	5.80	5.94	5.93

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b. Reverse. The test was repeated with the xxxx operating in the reverse direction. The results, displayed in MPH, are presented in Tables 4 through 6.

TABLE 4. REVERSE DIRECTION, GRAVEL

Selected Speed	Test Run Speed (mph)					Average Speed (mph)
	1	2	3	4	5	
XXXX	0.60	0.60	0.60	0.60	0.60	0.60
XXXX	2.36	2.41	2.36	2.40	2.41	2.39
XXXX	6.06	5.90	5.90	5.98	6.06	5.98

TABLE 5. REVERSE DIRECTION, GRASS

Selected Speed	Test Run Speed (mph)					Average Speed (mph)
	1	2	3	4	5	
XXXX	0.60	0.60	0.59	0.60	0.60	0.60
XXXX	2.38	2.36	2.33	2.41	2.40	2.38
XXXX	5.88	6.06	5.83	6.06	5.90	5.95

6.8.4 Braking Data Example.

TABLE 6. BRAKING DATA

METHOD OF BRAKE APPLICATION	ROAD SPEED, MPH	STOPPING DISTANCE	
		M	FT
DIRECT MODE			
Throttle release	5.6	1.1	3.8
Joystick bump	5.6	0.9	3.0
E-stop button	5.6 [a]	1.1	3.5
INDIRECT MODE			
Throttle release	3.6	2.0	6.5
	5.6	3.0	10.0
Joystick bump	3.6	0.4	1.3
	5.6	1.0	3.3
E-stop button	3.6	0.5	1.8
	5.6 [a]	0.9	3.0
a = Engine stalled			

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APPENDIX A. ABBREVIATIONS.

DoD	Department of Defense
FOV	Field of View
IR	Infrared
JAUS	Joint Architecture of Unmanned Systems
JGRE	Joint Ground Robotics Enterprise
OCU	Operator Control Unit
RF	Radio Frequency
SUT	System under test
TIRS	Test Incident Reports
TOP	Test Operations Procedure
UGV	Unmanned Ground Vehicle

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APPENDIX B. REFERENCES.

1. TOP 02-2-540, Testing of Unmanned Ground Vehicles, 12 February 2009.

2. TOP 02-2-542, Safe Operation of Weaponized Unmanned Ground Vehicle (UGV) Systems, 13 July 2010.

3. MIL-STD-882E, Department of Defense Standard Practice for System Safety, 11 May 2012.

4. Software System Safety Implementation Process and Tasks Supporting MIL-STD-882E, Revision: B, 14 March 2018.

5. TOP 03-2-812 with Change Notice 1, Field of Vision - Vehicles, 11 May 2017.

6. TOP 01-2-610, Human Factors Engineering, 15 May 1990.

7. TOP 02-2-546, Teleoperated Unmanned Ground Vehicle (UGV) Latency Measurements, 11 January 2017.

For information only (related publications).

a. Unmanned Systems Safety Guide for DoD Acquisition, First Edition, (Version .96), Jan 2007.

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APPENDIX C. APPROVAL AUTHORITY.

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WHITE SANDS MISSILE RANGE, NEW MEXICO 88002-5000

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SUBJECT: Test Operations Procedure 02-2-541A Safe Operation of Mobile Unmanned Ground Vehicle (UGV) Systems

1. Test Operations Procedure (TOP) 02-2-541A Safe Operation of Mobile Unmanned Ground Vehicle (UGV) Systems has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP describes the procedures for determining the mobility capabilities and characteristics of remotely and teleoperationally controlled Unmanned Ground Vehicle (UGVs). It focuses on the contribution of teleoperation to overall mobility and safe operation.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231 White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

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